

1. Different charts supported in Power BI

Power BI provides many types of charts and visuals. The commonly used ones are:

- **Bar Chart** – Compare different categories.
- **Column Chart** – Compare values across categories.
- **Line Chart** – Show trends or changes over time.
- **Area Chart** – Show trends over time with emphasis on magnitude.
- **Pie Chart** – Show parts of a whole.
- **Donut Chart** – Similar to a pie chart, showing proportions.
- **Scatter Chart** – Show the relationship between two numerical variables.
- **Combo Chart** – Combine column and line charts to compare two measures.
- **Funnel Chart** – Show stages in a process, such as a sales funnel.
- **Ribbon Chart** – Show changes in ranking across categories over time.
- **Waterfall Chart** – Show how positive and negative values contribute to a total.
- **Treemap** – Show hierarchical data and proportions using rectangles.
- **Gauge Chart** – Show progress toward a target.
- **Card** – Display a single important number, such as Total Sales.
- **Table** – Display detailed data in rows and columns.
- **Matrix** – Display data in a cross-tabular format with rows, columns, and hierarchy.
- **Map/Filled Map** – Show geographical data.

2. When to use which chart? With examples

Bar Chart

Use when: You want to compare different categories.

Example: Compare sales of different products.

Product A = ₹50,000

Product B = ₹70,000

Product C = ₹40,000

A Bar Chart makes it easy to see which product has the highest sales.

Column Chart

Use when: You want to compare categories vertically.

Example: Compare monthly sales for January, February, and March.

Line Chart

Use when: You want to show a trend over time.

Example: Show sales from January to June to see whether sales are increasing or decreasing.

Pie/Donut Chart

Use when: You want to show percentage or proportion of a total.

Example: Show what percentage of total sales comes from each product.

Scatter Chart

Use when: You want to find a relationship between two numerical values.

Example: Compare Advertising Cost vs Sales to see whether higher advertising spending is associated with higher sales.

Combo Chart

Use when: You want to compare two different measures together.

Example: Show Sales using columns and Profit using a line for each month.

Funnel Chart

Use when: You want to show the stages of a process.

Example:

Leads → Opportunities → Proposals → Sales

Waterfall Chart

Use when: You want to understand how different increases and decreases result in a final value.

Example:

Starting Revenue → Expenses → Tax → Final Profit.

Treemap

Use when: You want to show proportions across many categories or hierarchical data.

Example: Sales by Category → Product.

Gauge

Use when: You want to compare an actual value with a target.

Example:

Target Sales = ₹10 lakh

Actual Sales = ₹8 lakh

Card

Use when: You want to display one important KPI.

Example:

Total Sales = ₹25 lakh

3. Difference between Matrix Visual and Table Visual

Feature	Table Visual	Matrix Visual
Structure	Simple rows and columns	Rows and columns with hierarchy
Hierarchy	Limited	Supports hierarchy
Expand/Collapse	No	Yes
Grouping	Limited	Strong grouping capability
Subtotals	Limited	Supports subtotals
Drill Down	Limited	Supports hierarchical drill-down
Best for	Detailed records	Summarized and hierarchical data

Simple example

Table Visual:

Product	Region	Sales
Laptop	West	₹50,000
Mobile	West	₹30,000
Laptop	East	₹45,000

It mainly shows individual records.

Matrix Visual:

Region	Laptop	Mobile	Total
West	₹50,000	₹30,000	₹80,000
East	₹45,000	₹35,000	₹80,000
Total	₹95,000	₹65,000	₹1,60,000

A matrix can also have a hierarchy such as:

Region → State → City

You can expand and collapse each level.

- Easy way to remember

Table = Detailed data

Matrix = Summary + Hierarchy + Expand/Collapse